

Phy Sure Mark 30+mark

- **Define** Miller indices and draw the plane (001)(110)(101)(111). **05**
- **Show** that in a crystal of cubic structure, the distance between the planes with miller indices h,k,l is equal to $d = \frac{a}{\sqrt{h^2+k^2+l^2}}$. **05**
- **Or** Distinction between metal, insulator and semiconductor in terms of energy band.
- **Calculate** The following properties of ${}^4_2\text{He}$, ${}^{56}_{26}\text{Fe}$, ${}^{238}_{92}\text{U}$, ${}^{35}_{17}\text{Cl}$ **05-07**
*Nuclear Mass *Size *Density * Mass Defect *N.Charge *Binding.
- **Show** that the packing fraction for an fcc lattice is 0.74 write the meaning of this number. (Sc , **Bcc , Fcc**) **05**
- **Explain** The co-ordination number of **NaCl**, CsCl.
- **Define** Crystal systems, crystalline, non crystalline solid , lattice ,Basic lattic point , Unit cell , etc. **05**
- **Explain** “A moving clock always appers to go slow”. **Or** **05**
- **Derive** Einstine’s mass – energy relation. **05**
- **State** the laws of radioactive distegration and prove it **07**
- **Or** Haif Life / mean life of a radioactive. **05**
- **Math problem (10+ mark)**